

Denoising Diffusion Implicit Models

DDPM

$$q(x_t | x_0) = N(x_t; \sqrt{\alpha_t} x_0, (1 - \alpha_t) I) \quad \text{diffusion 的分布}$$

$$x_t = \sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \varepsilon \quad \varepsilon \sim N(0, I)$$

$$q(x_{t+1} | x_t) = \frac{q(x_t | x_{t-1}) q(x_{t+1})}{q(x_t)}$$

$$\textcircled{1} \hat{x}_0 = \frac{1}{\sqrt{\alpha_t}} x_t + \frac{\sqrt{1 - \alpha_t}}{\sqrt{\alpha_t}} \varepsilon_0$$

$$q(x_{t+1} | x_t, x_0) = \frac{q(x_t | x_{t-1}) q(x_{t+1} | x_0)}{q(x_t | x_0)} = N(x_{t+1}; u(x_t, x_0), \sigma_t^2 I)$$

reverse 的分布

$$u(x_t, x_0) = \frac{\sqrt{\alpha_t}(1 - \alpha_{t-1})}{1 - \alpha_t} x_t + \frac{\sqrt{\alpha_{t-1}} \beta_t}{1 - \alpha_t} x_0$$

$$\sigma_t^2 = \frac{1 - \alpha_{t-1}}{1 - \alpha_t} \beta_t$$

DDIM

已知: $q(x_t | x_0)$ $q(x_{t-1} | x_0)$ 未知 $q(x_t | x_{t-1})$

求 $q(x_{t-1} | x_t, x_0)$

$$\text{设 } x_{t-1} = m x_t + n x_0 + \sigma \varepsilon_1$$

$$\begin{cases} x_t = \sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \varepsilon_2 \\ x_{t-1} = \sqrt{\alpha_{t-1}} x_0 + \sqrt{1 - \alpha_{t-1}} \varepsilon_3 \end{cases}$$

$$x_{t-1} = m_t (\sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \varepsilon_2) + n_t x_0 + \sigma_t \varepsilon_1$$

$$= (\underline{m_t \sqrt{\alpha_t}} + n_t) x_0 + \underline{m_t \sqrt{1 - \alpha_t}} \varepsilon_2 + \underline{\sigma_t} \varepsilon_1$$

$$\Rightarrow \begin{cases} m_t \sqrt{\alpha_t} + n_t = \sqrt{\alpha_{t-1}} \end{cases}$$

$$\Rightarrow \begin{cases} m_t = \sqrt{\frac{1 - \alpha_{t-1} - \sigma_t^2}{1 - \alpha_t}} \end{cases}$$

$$m_t^2(1-\bar{\alpha}_t) + \sigma_t^2 = 1 - \bar{\alpha}_{t-1}$$

$$n_t = \sqrt{\bar{\alpha}_{t-1}} - \sqrt{\frac{\bar{\alpha}_t}{1-\bar{\alpha}_t}}(1-\bar{\alpha}_{t-1}-\sigma_t^2)$$

代回 m_t 、 n_t

$$X_{t+1} = \sqrt{\frac{1-\bar{\alpha}_{t+1}-\sigma_{t+1}^2}{1-\bar{\alpha}_t}} X_t + \left(\sqrt{\bar{\alpha}_{t+1}} - \sqrt{\frac{\bar{\alpha}_t}{1-\bar{\alpha}_t}}(1-\bar{\alpha}_{t+1}-\sigma_{t+1}^2) \right) \cdot X_0 + \sigma_{t+1} \varepsilon$$

$$= \sqrt{\bar{\alpha}_{t+1}} X_0 + \sqrt{1-\bar{\alpha}_{t+1}-\sigma_{t+1}^2} \left(\frac{1}{\sqrt{1-\bar{\alpha}_t}} X_t - \frac{\sqrt{\bar{\alpha}_t}}{\sqrt{1-\bar{\alpha}_t}} X_0 \right) + \sigma_{t+1} \varepsilon$$

DDIM
采样公式

$$X_{t+1} = \sqrt{\bar{\alpha}_{t+1}} \underbrace{X_0}_{\hat{X}_{0|t}} + \sqrt{1-\bar{\alpha}_{t+1}-\sigma_{t+1}^2} \underbrace{\frac{X_t - \sqrt{\bar{\alpha}_t} X_0}{\sqrt{1-\bar{\alpha}_t}}}_{\Sigma_0(X_t, t)} + \sigma_{t+1} \varepsilon$$

$q(X_s | X_T)$ 可实现跳步的采样, 以实现加速

DDPM $T=1000 \ 999 \ 998 \ \dots \ 3 \ 2 \ 1 \ 0$

DDIM $= T=1000 \dots 888 \dots 777 \dots 666 \dots 111 \ 0$ 无视 $q(X_t | X_{t-1})$ 这一条件

DDIM 训练目标

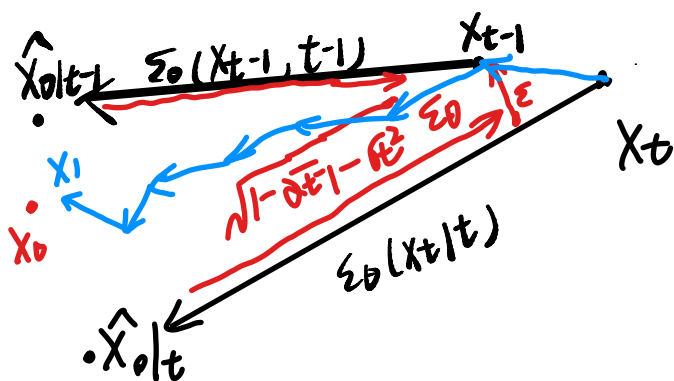
$$\begin{array}{ccccccc} \text{DDPM} & \bar{\alpha}_1 & \bar{\alpha}_2 & & \bar{\alpha}_t & \dots & \bar{\alpha}_{T-1} & \bar{\alpha}_T \\ & & & & \downarrow & & & \\ & & & & X_0 \rightarrow X_t & & & \\ & & & & \downarrow & & & \\ & & & & \Sigma_0(X_t, t) & & & \end{array}$$

DDIM
挑选了DDPM一个子集

$$\begin{array}{ccccccc} & \bar{\alpha}_{100} & \bar{\alpha}_{200} & \dots & \bar{\alpha}_{900} & & \bar{\alpha}_{1000} = \bar{\alpha}_T \\ & & \downarrow & & & & \\ & & \Sigma_0(X_{100}, 200) & & & & \Sigma_0(X_T, T) \end{array}$$

图示理解:

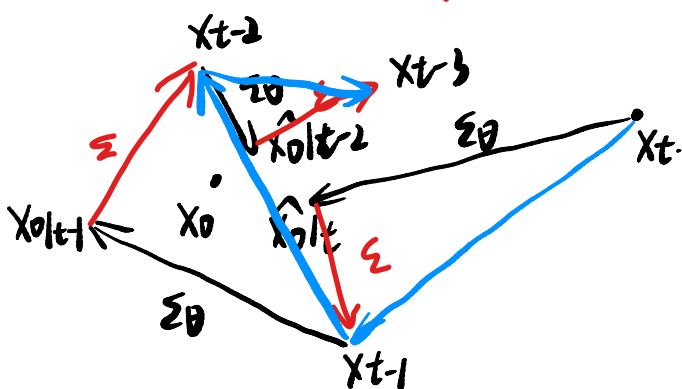
$\sigma_t \neq 0$



case 1: $\sigma_t = 0$ (退化为 DDPM) 少一些随机性

case 2: $\sqrt{1-\alpha_{t-1}-\sigma_t^2} = 0$

波动性很强, 看起来很乱



常微分方程

$$X_{t-1} = \sqrt{\alpha_{t-1}} X_0 + \sqrt{1-\alpha_{t-1}-\sigma_t^2} \epsilon_0(X_t, t) + \sigma_t \epsilon$$

$$\sigma_t = 0 \quad X_{t-1} = \sqrt{\alpha_{t-1}} \frac{1}{\sqrt{\alpha_t}} (X_t - \sqrt{1-\alpha_t} \epsilon_0(X_t, t)) + \sqrt{1-\alpha_{t-1}} \epsilon_0(X_t, t)$$

$$\frac{X_{t-1}}{\sqrt{\alpha_{t-1}}} = \frac{X_t}{\sqrt{\alpha_t}} - \sqrt{\frac{1-\alpha_t}{\alpha_t}} \epsilon_0(X_t, t) + \sqrt{\frac{1-\alpha_{t-1}}{\alpha_{t-1}}} \epsilon_0(X_t, t)$$

$$\frac{X_t}{\sqrt{\alpha_t}} - \frac{X_{t-1}}{\sqrt{\alpha_{t-1}}} = \left(\sqrt{\frac{1-\alpha_t}{\alpha_t}} - \sqrt{\frac{1-\alpha_{t-1}}{\alpha_{t-1}}} \right) \epsilon_0(X_t, t) \quad \begin{cases} a = \sqrt{\alpha} \\ \sigma = \sqrt{1-\alpha} \end{cases}$$

令自变量为 s

$$\frac{d}{ds} \left(\frac{X(s)}{a(s)} \right) = \frac{d}{ds} b(s) \cdot \epsilon_0(X_t, t) \quad s \in [0, 1]$$

已知 $X(1) \sim N(0, I)$ 求 X_0

DDPM/DDIM



$$X_T = X(1) = \epsilon \sim N(0, I)$$